THE NOVA RESEARCH CURRICULUM

Chemistry

Grade 8 · Term 2

Student edition

Outcomes: CH-8.2 balancing simple equations, reaction types, exothermic and endothermic reactions; CH-8.3 ionic and covalent bonds with dot-and-cross diagrams.

Lesson notes

- **Balancing equations (Weeks 1–4):** atoms are not created or destroyed, so both sides must have the same number of each atom. Change numbers in front, never the small numbers: $H_2 + O_2 \rightarrow H_2O$ becomes $2H_2 + O_2 \rightarrow 2H_2O$.
- **Reaction types (Weeks 5–8):** combination ($2Mg + O_2 \rightarrow 2MgO$), decomposition ($CaCO_3 \rightarrow CaO + CO_2$ on heating), displacement ($Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$), combustion. Exothermic reactions give out heat (burning, hand warmers); endothermic reactions take in heat (cold packs, photosynthesis). Measure temperature change.
- **Bonding (Weeks 9–12):** ionic bond: a metal gives electrons to a non-metal, making ions that attract (Na^+ Cl^-). Covalent bond: non-metals share pairs of electrons (H_2, H_2O, CH_4). Dot-and-cross diagrams.

  ▶ **Try it on NovaSchool:** Ionic and covalent bonds — open **Simulations** on your dashboard.

Workbook pages

Page	Task
1–2	Balance 12 equations.
3	Name the reaction type for 10 equations; temperature-change practical.
4	Dot-and-cross diagrams for NaCl, MgO, H_2, H_2O, CH_4.

Question bank

#	Outcome	Question	Your answer
1	CH-8.2	Balance: $__H_2 + O_2 \rightarrow __H_2O$	1 and 1 / 2 and 2 / 2 and 1 / 1 and 2
2	CH-8.2	$Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$ is a … reaction.	decomposition / displacement / combustion / neutralisation
3	CH-8.2	A reaction that gives out heat is …	endothermic / exothermic / physical / neutral
4	CH-8.2	Heating calcium carbonate to make calcium oxide and carbon dioxide is decomposition. True or false.	True / False

#	Outcome	Question	Your answer
5	CH-8.3	In an ionic bond, electrons are …	shared / transferred / destroyed / doubled
6	CH-8.3	Which has covalent bonds?	NaCl / MgO / H_2O / KCl
7	CH-8.3	Sodium forms an ion with charge …	−1 / +1 / +2 / 0
8	CH-8.3	Why do non-metals share electrons?	_____